Arda Kayaalp

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Personal Information

Birth: July 8, 1997, Turkey
Gender: Male
Citizenship: Turkish

Education

Middle East Technical University — **Mar. 2021 – Present**
M.Sc. Student at Department of Physics (3.36/4.00 CGPA) — *Ankara, Turkey*
Thesis Title: Application of the Quantal Diffusion Approach Based on the Stochastic Mean-Field Theory

Hacettepe University — **Aug. 2015 – Jun. 2019**
B.Sc. in Nuclear Engineering (Honours, 3.22/4.00 CGPA) — *Ankara, Turkey*
Graduation Project : A Study of $p - B^{11}$ *Fusion Reaction using Geant4 Toolkit*

Employment & Research Experience

R&D Engineer — **Jul. 2022 – May 2023**
Oncotech Medical Systems — *Ankara, Turkey*

- Research and algorithm development on medical data and image processing systems.
- Development of MATLAB-based prototype of a DICOM-RT viewer application with GUI.
- Research and algorithm development on a low-cost, fast and accurate optical surface tracking system for SGRT applications.
- Development of a low-cost prototype application of a SGRT system in C++ using OpenCV and CUDA toolkits.

Scientific Project Specialist — **Jan. 2020 – Jun. 2022**
Research and Application Center for Space and Accelerator Technologies at METU — *Ankara, Turkey*

- Magnetic alignment calculations of quadrupole magnets on METU-DBL.
- Proton beam parameter calculations and beam preparations for the irradiation tests.
- Periodic maintenance of the beamline elements on the METU-DBL.
- Monte Carlo simulations and radiation dose calculations for the irradiation tests.
- Research and analysis on chaotic and quantum random systems.
- Development of image processing applications to extract information from chaotic systems.

Undergraduate Researcher — **Dec. 2018 – Jan. 2020**
Research and Application Center for Space and Accelerator Technologies at METU — *Ankara, Turkey*

- Development of an image processing software with MATLAB that calculates the approximate area and centroid of the proton beam image on the beam screen at METU-DBL.
- Design and implementation of irradiation tests on METU-DBL.
- Basic proton beamline calculations with TURTLE Program.

**METU-DBL : Middle East Technical University Defocusing Beam Line*

Research Background

Multinucleon transfer mechanisms, computational radiation transport, experimental nuclear physics.

Internships

Çekmece Nuclear Research and Education Center — **Jun. – Jul. 2018**
Studied reactor physics, radiation protection and radioactive waste management. — *Istanbul, Turkey*

OMV Samsun Thermal Power Plant — **Jul. – Aug. 2017**
Studied thermodynamical aspects of a power plant. — *Samsun, Turkey*

Awards & Grants

The Scientific and Technical Research Council of Turkey (TUBITAK) **Mar. 2022 – Present**
TUBITAK Graduate Scholarship *Ankara, Turkey*

Random Power Hackathon, organised by randompower.eu & CERN Openlab **Oct. 2020**
Third Place as Team Hexagon *Virtual Event*
Project : Random Power generated TRNs for Geant4 HEP event simulations

Technical Skills

Hardware Experience : I possess fundamental skills in electronic laboratory work and soldering, along with hands-on experience in radiation detector electronics and radiation counting statistics. Furthermore, I have a basic understanding of ARM-based Linux development and proficiency in edge computing, making use of platforms such as Jetson Nano and Raspberry Pi.

Software Experience :

- **MATLAB:** Moderate to advanced knowledge. I modelled radiation shielding problems and developed detector readout interfaces and data and image processing applications. Additionally, I developed several GUI-based prototype applications.
- **Python:** Intermediate knowledge. I am generally using it for data processing and visualization.
- **C++:** Intermediate knowledge. I used it to develop Geant4-based particle simulations, and I am currently utilizing it for nuclear evaporation calculations.
- **Fortran:** Basic knowledge. I am using it for simple manipulations on legacy physics codes.
- **CMAKE:** Basic knowledge. I am using it to build cross-platform C++ applications.
- **Git:** Basic knowledge. I am using it to version local projects.
- **MCNP & SERPENT:** Basic to intermediate knowledge. Using SERPENT, I calculated the reactivity of LWR fuel rod geometry and modelled fuel burn-up cycles. With MCNP, I modeled radiation shielding geometries.
- **Geant4:** Basic to intermediate knowledge. I modelled simple nuclear interactions and radiation transport problems.
- **Linux:** Intermediate knowledge. I am using Linux as a development OS for all my projects.
- **LaTeX:** Intermediate knowledge. I am using it for academic papers and other documentation.

Language Proficiency

- Turkish - Native
- English - TOEFL iBT Score : 104 (C1)

Teaching Experience

I am occasionally tutoring undergraduate students on applications of nuclear physics courses.

Extracurricular Activities

Digital photography, charcoal drawing, fountain pen collecting.

Published Works

1. Kayaalp, A. *et al.* A quantal diffusion approach for multinucleon transfer in heavy-ion collisions. *Nuclear Physics A,* 122916 (2024)
2. Kayaalp, A. *et al.* A theoretical study on quasifission and fusion–fission processes in heavy-ion collisions. *The European Physical Journal A* **60,** 79 (2024)

References

- **Prof. Osman Yılmaz - Academic Advisor**
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- **Assoc. Prof. Erol Çubukçu - B.Sc. Graduation Project Advisor**
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- **Çağrı Yazgan - Nuclear Engineer - R&D Coord. at Oncotech Medical Systems**
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